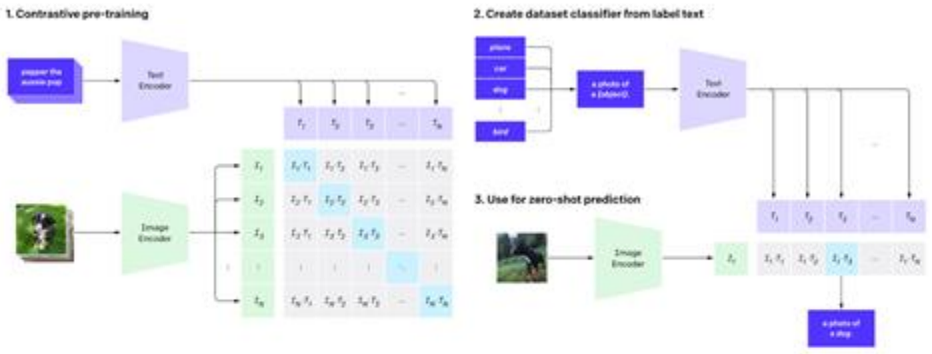
Open World Image Semantic Segmentation

- Introduction to **CLIP** (Contrastive Language-Image Pre-training)
- Aligns the CLS token of image and text encoder via InfoNCE
- Zero-shot image classification



$$\mathcal{L}_{\mathbf{x}} = \frac{1}{k}\sum_{i=1}^{k}\left(\frac{e^{\phi(\mathbf{x}_i,\mathbf{y}_i)}}{\sum_{j=1}^{k} e^{\phi(\mathbf{x}_i,\mathbf{y}_j)}}\right)$$

$$\mathcal{L}_{\mathbf{y}} = \frac{1}{k}\sum_{i=1}^{k}\left(\frac{e^{\phi(\mathbf{x}_i,\mathbf{y}_i)}}{\sum_{j=1}^{k} e^{\phi(\mathbf{x}_j,\mathbf{y}_i)}}\right)$$

$$f_v : \mathbb{R}^{C,H,W} \rightarrow \mathbb{R}^{D_v} \text{ and } f_t : \mathbb{R}^{l} \rightarrow \mathbb{R}^{D_t}$$

$$e_v : \mathbb{R}^{D_v} \rightarrow \mathbb{R}^{D} \text{ and } e_t : \mathbb{R}^{D_t} \rightarrow \mathbb{R}^{D}$$

$$\phi(\mathbf{x},\mathbf{y}) = \left(\frac{e_v(f_v(\mathbf{x}))}{|e_v(f_v(\mathbf{x}))|} \cdot \frac{e_t(f_t(\mathbf{y}))}{|e_t(f_t(\mathbf{y}))|}\right)$$

$$\mathcal{L}_{\text{InfoNCE}} = 1/2(\mathcal{L}_{\mathbf{x}} + \mathcal{L}_{\mathbf{y}})$$

Attentive Mask CLIP

A-CLIP

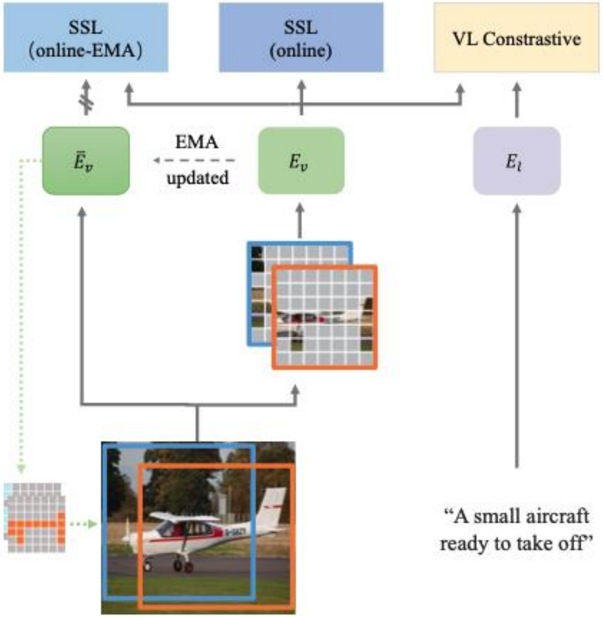
- Main idea:
  - Drop irrelevant tokens!!
  - Drop almost >= 50%
- Random dropping degrades performance though
  - Masked Image Modeling (MIM) works with random dropping but not CLIP, why?

Attentive Mask

$$s_P = \frac{1}{HL}\sum_{l=1}^{L}\sum_{h=1}^{H}\text{Softmax}\left(\frac{\mathbf{f}_{lh}^{q}(CLS)\cdot\mathbf{f}_{lh}^{k}(P)}{\sqrt{C}}\right)$$

where l denotes the layer index; h denotes the attention head index; $\mathbf{f}_{lh}^{q}(CLS)$ denotes the query embedding of the $[CLS]$ token at Layer l and Head h; $\mathbf{f}_{lh}^{k}(P)$ denotes the key embedding of Layer l and Head h for an image token at location P; C is the number of channels for the query and key embedding.

- EMA for attention score computation
  - A-CLIP-eff: half-res for speed as it does not need to be very accurate
- Shared EMA scored map for multiple views, efficiency
- Multiple masked views
  - Auxiliary contrastive learning
    - Between a masked view and EMA feature (BYOL)
    - Between masked views (SIMCLR)



Sigmoid Loss based CLIP

SGLIP

- Recall:

$$\mathcal{L}_{\mathbf{x}} = \frac{1}{k}\sum_{i=1}^{k}\left(\frac{e^{\phi(\mathbf{x_i},\mathbf{y_i})}}{\sum_{j=1}^{k} e^{\phi(\mathbf{x_i},\mathbf{y_j})}}\right)$$

$$\mathcal{L}_{\mathbf{y}} = \frac{1}{k}\sum_{i=1}^{k}\left(\frac{e^{\phi(\mathbf{x_i},\mathbf{y_i})}}{\sum_{j=1}^{k} e^{\phi(\mathbf{x_j},\mathbf{y_i})}}\right)$$

Inefficient and prevents large batch size

SGLIP

- Instead of global normalization as in CLIP, separate pair processing
- A lot of negative pairs causing a lot of imbalance during optimization
  - Massive over-correction in the training results
- Added a bias term and temperature
- Z_{ij} is the label (1 for positive pair and –1 otherwise)
- Batch size can go up to a million, but typically 32k

$$-\frac{1}{|\mathcal{B}|}\sum_{i=1}^{|\mathcal{B}|}\sum_{j=1}^{|\mathcal{B}|}\underbrace{\log\frac{1}{1+e^{z_{ij}(-t\mathbf{x}_i\cdot\mathbf{y}_j+b)}}}_{\mathcal{L}_{ij}}$$

Allows for chunking

Attendance

Assignment Due 9/24

- Dataset: Human Action Recognition (HAR)
  - Same train/test split
- Fine tune on HAR and compare CLIP, A-CLIP, A-CLIP* and SGLIP
  - A-CLIP* does not use the attentive mask, but simply random token dropping
- Report performance, speed and memory consumption
  - SGLIP is able to load very large batch size, use 32k
  - For CLIP, A-CLIP (4096)
- Report, code, video to be submitted

Team Project Presentations

- 10 points for division of tasks. A team member doing the same thing as another team member will split the 10 points. So if two read Dreamfusion paper as an example, each gets 5 points. If four read the same paper as an example, each gets 2.5 points.
  - Therefore it is best to have clear division of tasks, and each team member showcase the contributions made.
- 5 points for contributions to to the final goal. If the final goal is unclear, this will get close to 0. If the final goal is clear, this will be assessed based on the contributions of your task.
- 5 points for execution. If your task is say object detection in NFL games and you actually ran a Yolo-World for example, this is good. If you only read a paper for your task and no execution, 0 point on this.